

ASHEVILLE 8 SSW, NC, USA

WMO: 746960

Lat: 35.4945N

Lon: 82.6142W

Elev: 656

StdP: 93.69

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53877

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.2	-16.7	0.9	-7.1	-14.5	1.2	-4.9	2.2	2.4	2.2	2.4	0.6	N/A	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	30.1	21.2	29.0	21.0	27.9	20.7	23.3	27.1	22.7	26.6	22.1	25.9	0.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	18.3	24.9	21.5	17.5	24.5	20.9	16.8	24.0	72.8	27.2	70.5	26.7	68.2	26.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 2.2	2.0	1.8	-15.0	32.1	2.7	1.6	-16.9	33.3	-18.5	34.2	-20.0	35.1	-21.9	36.3		
(5)			-15.5	24.3	2.8	0.7	-17.5	24.8	-19.1	25.2	-20.7	25.7	-22.7	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	2.1	4.1	8.1	12.7	16.7	20.5	21.9	21.5	18.5	12.9	7.5	4.3		
	DBStd	7.95	5.40	4.75	4.74	3.99	3.40	2.17	1.91	2.03	3.00	4.19	4.37	4.75		
	HDD10.0	842	248	173	96	18	2	0	0	0	1	19	98	186		
	HDD18.3	2436	502	399	318	171	68	7	2	3	33	172	324	435		
	CDD10.0	1797	4	7	35	101	211	314	367	357	257	111	24	9		
	CDD18.3	351	0	0	0	3	18	71	111	102	39	6	1	0		
	CDH23.3	2582	0	1	8	87	228	557	732	645	283	38	2	0		
	CDH26.7	534	0	0	0	9	29	122	175	144	50	4	0	0		
(14) Wind	WSAvg	0.5	0.7	0.8	0.8	0.8	0.5	0.4	0.3	0.3	0.3	0.5	0.6	0.6		
(15) Precipitation	PrecAvg	1498	125	116	137	120	109	141	139	143	131	96	107	135		
	PrecMax	2150	297	237	256	221	253	269	361	256	491	223	219	259		
	PrecMin	1008	33	49	51	54	37	40	49	26	24	7	38	59		
	PrecStd	257	63	46	49	41	54	56	69	57	90	57	46	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	21.9	24.6	28.1	29.2	31.3	31.4	31.4	30.0	26.8	23.1	20.1		
		MCWB	13.0	14.3	14.1	16.0	18.8	20.7	22.1	21.5	20.1	19.6	15.5	14.1		
	2%	DB	16.2	18.0	22.0	25.8	27.4	29.5	29.9	29.8	28.1	24.2	20.2	17.2		
		MCWB	11.7	11.8	13.6	15.6	18.5	20.8	21.8	21.8	20.0	16.9	13.3	13.3		
	5%	DB	13.8	15.6	19.7	23.9	26.0	28.0	28.7	28.3	26.6	22.2	18.1	15.0		
		MCWB	10.0	10.0	12.3	14.7	18.1	20.4	21.7	21.5	19.8	16.6	13.2	11.6		
	10%	DB	11.1	12.9	17.3	21.9	24.4	26.6	27.4	26.9	24.9	20.4	16.2	12.6		
		MCWB	8.6	8.2	11.0	14.0	17.6	20.0	21.5	21.2	19.3	15.9	11.9	9.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.8	16.0	18.5	21.3	23.2	24.4	24.4	22.4	21.3	18.1	16.2		
		MCDB	17.9	19.9	21.4	24.2	25.5	27.0	27.9	27.7	26.3	24.6	19.7	18.3		
	2%	WB	12.9	13.8	14.9	17.3	20.3	22.4	23.3	23.4	21.6	19.6	16.0	14.3		
		MCDB	15.3	16.9	19.8	22.9	24.6	26.5	27.2	27.1	25.6	22.4	18.4	16.4		
	5%	WB	10.8	11.2	13.7	16.2	19.6	21.8	22.7	22.7	20.8	18.3	14.6	12.2		
		MCDB	12.4	14.1	18.0	21.2	23.8	25.9	26.7	26.4	24.3	20.7	17.0	14.2		
	10%	WB	8.8	9.0	12.2	15.2	18.7	21.0	22.2	22.0	20.1	16.9	12.7	10.2		
		MCDB	10.7	12.0	15.9	19.9	22.8	25.1	26.0	25.6	23.5	19.3	15.2	12.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.4	13.2	14.4	13.2	12.4	11.3	11.2	11.6	13.0	13.9	11.6		
		MCDBR	16.4	17.3	17.7	18.3	15.8	14.5	13.1	13.4	14.3	15.2	16.9	14.3		
	5% WB	MCWBR	11.2	11.1	9.9	9.2	7.9	7.0	6.2	6.4	7.0	8.5	10.1	9.8		
		MCWBR	12.4	13.8	14.3	14.5	12.3	12.2	11.4	11.1	10.8	10.2	11.2	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.303	0.332	0.372	0.426	0.451	0.492	0.490	0.385	0.351	0.313	0.291		
	taud		2.545	2.513	2.457	2.346	2.210	2.172	2.087	2.072	2.352	2.442	2.516	2.581		
	Ebn at Noon		917	946	939	911	862	837	800	793	871	875	883	900		
	Edh at Noon		79	91	105	124	144	149	161	161	115	97	80	72		
(44) All-Sky Solar Radiation	RadAvg		2.57	3.20	4.24	5.25	5.77	6.07	5.65	5.30	4.59	3.83	2.88	2.22		
	RadStd		0.16	0.34	0.27	0.40	0.42	0.42	0.40	0.25	0.41	0.39	0.25	0.20		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +139	(m) N/A	(46)
(47)	Regional (40 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+100	(47)

Nomenclature: See separate page